Embedded Systems Development

Exercise 6

1. Design an "auto fan" using ATmega328P microcontroller and give your design in a schematic diagram.

   The "auto fan" is an embedded system that drives the 3 – 6 V DC motor of a fan by sensing the room temperature. It captures the room temperature using an LM35 temperature sensor and automatically drives the motor at full speed when the room temperature is 35 °C or higher and turns the motor off when the room temperature is 25 °C or lower. As the room temperature rises or falls between 25 °C and 35 °C, the "auto fan" smoothly increases or decreases the speed of the motor, respectively. The "auto fan" demonstrates how a physical phenomenon is sensed and converted into an analog voltage using a transducer, how the analog voltage is converted into a discrete digital value that can be processed by a CPU, and how the processed value is then used to generate an analog-like output using digital means to drive an actuator.

   Schematic Diagram

2. Write the required firmware for the "auto fan" using the wrapper functions in the Arduino IDE.

3. Write the required firmware for the "auto fan" in C using the Arduino IDE.
   Note: You are required to use Arduino IDE instead of Microchip Studio IDE and write your code in C programming language without using any wrapper functions available in Arduino platform.